Online Identification of Environment Hunt–Crossley Models Using Polynomial Linearization

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***Abstract*—Online environment dynamic estimates are often used for the control of robots, telerobots, and haptic systems. The nonlinear Hunt–Crossley (HC) model, which is physically consistent with the behavior of soft objects with limited deformation at a single point of contact, is being increasingly used in robotic control systems. The HC model can be identified online using a single-stage log linearization technique; however, the accuracy and applicability of the existing method is limited. We propose a two-stage polynomial identification method, which uses a quadratic approximation in the first stage to generate a linearly parameterized model of the HC dynamics (Quad-Poly). The coefficients of the Quad-Poly model are then used in the second stage to extract the HC parameters using a lookup table and recursive least squares parameter estimation. The proposed method is experimentally assessed against a previous natural logarithm linearization method, and further tested for time-varying environment dynamics and human-generated trajectories and for robustness against uncertainties in the measured data and system parameters.**

***Index Terms*—Hunt–Crossley (HC), Kelvin Voight (KV), online model identification, parameter estimation, polynomial linearization.**

I. Introduction

ESTIMATED models of physical objects (or environments) are commonly used to control the interaction between a robot and its surroundings. Models can be used to predict the amount of force that will be exerted by the environment, or to identify the environment that is in contact with the robot. Environment models can be estimated online from position and force data as the robot moves along a sufficiently rich excitation trajectory [1], [2]. Alternatively, the dynamics can be estimated from force data alone through the use of active observers that estimate stiffness from the control command [3]–[5]. The modeled dynamics can be used to improve force control [3] or in techniques such as impedance control, which regulates position and contact force simultaneously [1], [6]. In telerobotic systems, impedance reflecting and model-mediated bilateral controllers transmit an online estimate of the environment dynamic model to the master side for improved transparency [7]–[10]. Offline environment estimates have also been used for haptic augmentation of physical environments [11], and to render virtual environments for a master robot controlling a semiautonomous slave [12].

The Kelvin Voight (KV) linear spring-damper model for a point of contact is given by

$$F = \begin{cases} \bar{K}x + \bar{B}\dot{x} & : x >= 0 \\ 0 & : x < 0 \end{cases} \tag{1}$$

where x and $\dot{x}$ denote position and velocity, and $\bar{K}$, $\bar{B}$ are the environment stiffness and damping. KV is commonly used in robotic systems and is easily identified online using recursive least squares (RLS) [1], [2]. More generally, locally-linear mass-spring-damper systems can be used to model the haptic properties of nonlinear objects such as knobs and switches [13], [14]. Although well-suited for stiff environments, KV is physically inconsistent with soft environments [15].

1) When the robot first makes contact with the environment, the positive velocity causes an unrealistic discontinuous force. This force is viscous in nature, does not depend on environment stiffness, and is not intended to model the impact impulse forces.
2) As the robot breaks contact with the environment, the negative velocity causes sticky forces, pulling the robot back inside the environment.
3) The kinetic energy conservation during a collision, modeled by the coefficient of restitution, is incorrectly predicted to be independent of the initial velocity.

A nonlinear model, proposed by Hunt and Crossley, removes the discontinuity effects of the KV model by making the damping term position dependent [15], [16]. The Hunt–Crossley (HC) model for a single point of contact is

$$F = \begin{cases} Kx^n + Bx^n\dot{x} & : x >= 0 \\ 0 & : x < 0 \end{cases} \tag{2}$$

where n is a unit-less constant typically between 1 and 2, which reflects the material and the geometry of the contact surface [17], [18]. The coefficient of restitution derived from the HC model has also been shown to be dependent on the initial contact velocity [17]. Note that K and B in the HC model (2) are not analogous to $\bar{K}$ and $\bar{B}$ in the KV model (1). However, one can consider Kx^{n-1} and Bx^n as position-dependent stiffness and damping.

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Although harder to identify, HC models are more accurate for soft materials such as heart and breast phantom tissues [19], [20]. The HC model can be used to represent the contact dynamics of environments with simple geometry under the assumptions that the environment deformation is small, and that there is a single point of contact between the tool and environment. More complex models, such as finite-element, shall be utilized for environments experiencing significant deformations or distributed contact forces [21], [22].

The HC model is increasingly being adopted for the design of robotic controllers for interacting with soft environments. Early examples include the incorporation of HC dynamics into model-mediated telerobotic feedback [23] and the use of the HC model to design neural network based adaptive robotic controllers [24]. The stability of a robot coupled with an HC environment has been theoretically analyzed and experimentally tested in [25] with the aim of guaranteeing safe human–robot interaction. HC models of soft tissues have been used in surgical telerobotics [26] and state feedback force control for minimally invasive surgery [27]. These controllers, however, require *a priori* the knowledge of the model parameter estimates $[\hat{K}, \hat{B}, \hat{n}]$ for accurate position tracking. Offline HC parameter estimates have also been used to render virtual tumors for haptically augmented breast tissue palpation [20].

The consistency and mathematical simplicity of HC makes it a suitable candidate for robotic control applications, however, online HC parameter estimation poses a challenge. Since HC cannot be linearly parameterized, RLS cannot be directly used to identify $[\hat{K}, \hat{B}, \hat{n}]$. Instead, the two-stage estimation method or the linearized logarithmic identification (LLI) method must be used [18], [28]. These methods are restricted by the choice of initial parameters, the damping coefficient B and poor performance near the environment boundary.

Recently, the authors proposed the use of polynomial identification (PI) for the online estimation of HC models [29]. They obtained the so-called quadratic-polynomial (Quad-Poly) model by replacing the nonlinear term x^n with a Quad-Poly, and identified the Quad-Poly parameters using RLS. The HC parameters $[\hat{K}, \hat{B}, \hat{n}]$ were then extracted from the Quad-Poly estimate using lookup tables. The proposed methodology was evaluated in simulations against the LLI method.

In this paper, we propose a new two-stage polynomial identification (TSPI) method, in which the HC parameters $[\hat{K}, \hat{B}]$ are estimated in a second-stage RLS instead of directly from the Quad-Poly estimate. Quad-Poly and TSPI are experimentally evaluated on three types of environments, and shown to be more accurate and less restrictive than LLI. We will also rigorously analyze the robustness of the proposed methods against force noise and bias, contact point estimation error, velocity filter bandwidth, and uncertainty in the maximum penetration depth. The increased accuracy of HC models is expected to lead to better performance in robotic contact tasks.

Remark: Throughout the paper N and cm are used for force and length. Likewise [16], [18], [26]–[28], we do not consider units for K and B in the HC model, since the units will be a function of $\hat{n}$, which changes during the identification and its final value bears some degree of uncertainty.

II. Previous HC Identification Methods

Typical robotic environment identification uses data collected by position and force sensors as the robot moves along an excitation trajectory to develop a contact impedance model. Although force sensors are difficult to use in some small-scale applications such as telesurgery [26], they are easily incorporated in larger scale applications such as robot–human interaction [25], [30]. In situations where a force sensor is difficult to install, a neural-network force observer can alternatively be used [31].

The first online HC identification method proposed by Diolaiti *et al.* linearized the HC equation by using two separate RLS stages for $\hat{n}$ and $[\hat{K}, \hat{B}]$. However, the interconnected feedback structure severely limited the convergence time and required that the initial estimates be close to their final values [28]. To achieve faster convergence independently of the initial conditions, Haddadi and Hashtrudi-Zaad proposed the LLI method, which used a single-stage linearized RLS estimator [28]. LLI identification was validated experimentally in [28] and proved to be a significant improvement over [18]. LLI has been evaluated for in vivo modeling of abdomen tissue [32] and used in various robotic and haptic applications [11], [20], [23], [26]. LLI linearizes the HC dynamics by taking the natural logarithm of (2) [28]. In order to do so, the following two assumptions must be made:

$$||\dot{x}||_\infty < \frac{0.1K}{B} \tag{3}$$

$$\frac{\epsilon}{Kx^n(t)} \ll 1 \tag{4}$$

where ϵ is the combined measurement noise and modeling error, and is assumed to be a zero-mean stochastic process in order for the estimate to be unbiased. (3) restricts the robot's velocity for a given stiffness to damping ratio, and (4) is violated near the environment boundary. In order to avoid singularities, F and x must be greater than zero. The HC dynamics can then be represented by

$$\ln(F) \approx \ln(K) + \frac{B}{K}\dot{x} + n\ln(x) + \frac{\epsilon}{Kx^n} \tag{5}$$

which is linearly parametrized as

$$y_k = \boldsymbol{\phi}_k^T \boldsymbol{\theta}_k + \epsilon_k, \quad x_k > 0 \tag{6}$$

where k is the discrete time index, y_k is the system output, $\boldsymbol{\theta}_k$ is a vector containing the dynamic parameters, and $\boldsymbol{\phi}_k^T$ is the regressor vector, given by

$$\boldsymbol{\phi}_k^T = [1, \dot{x}_k, \ln(x_k)],\ \boldsymbol{\theta}_k = \left[\ln(K_k), \frac{B_k}{K_k}, n_k\right]^T,\ y_k = \ln(F_k). \tag{7}$$

The exponentially weighted recursive least squares (EWRLS) algorithm is chosen for parameter estimation since it has a faster convergence time than RLS. EWRLS includes a "forgetting factor," which causes the estimation to disregard older data and put weight on new ones. From [33], the EWRLS update

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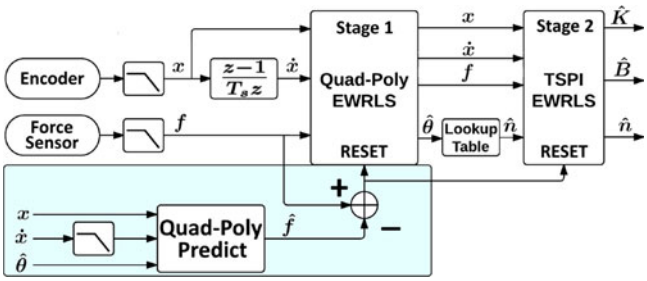


Fig. 1. Block diagram demonstrating the proposed two-stage polynomial estimation process, including digital filter locations and reset mechanism for parameter tracking.

equations are

$$\boldsymbol{L}_{k+1} = \frac{\boldsymbol{P}_k \boldsymbol{\phi}_{k+1}}{\lambda + \boldsymbol{\phi}_{k+1}^T \boldsymbol{P}_k \boldsymbol{\phi}_{k+1}} \tag{8}$$

$$\boldsymbol{P}_{k+1} = \frac{1}{\lambda}\left[\boldsymbol{P}_k - \boldsymbol{L}_{k+1}\boldsymbol{\phi}_{k+1}^T \boldsymbol{P}_k\right] \tag{9}$$

$$\hat{\boldsymbol{\theta}}_{k+1} = \hat{\boldsymbol{\theta}}_k + \boldsymbol{L}_{k+1}\left[y_{k+1} - \boldsymbol{\phi}_{k+1}^T \hat{\boldsymbol{\theta}}_k\right] \tag{10}$$

where $\boldsymbol{P}$ is the covariance matrix, $\boldsymbol{L}$ is the adaption gain, and λ is a forgetting factor typically between 0.9 and 1, $\boldsymbol{P}_0$ is chosen to be the identity matrix, and in this paper $\hat{\boldsymbol{\theta}}_0 = [0\ 0\ 0]^T$. If $\lambda < 1$, EWRLS is more sensitive to new inputs, and thus, reacts faster to stimuli than when $\lambda = 1$, but is less stable steady-state estimate. In order to achieve a balance, an exponential forgetting factor with a time-varying λ is used

$$\lambda(t) = 1 - 0.01e^{-t/\mu}. \tag{11}$$

For the experiments in this paper, $\mu = 0.5$ s is selected. The above forgetting factor is implemented in discrete time for a sampling rate of 1 kHz as

$$\lambda_k = 1 - \gamma_k \tag{12}$$

$$\gamma_k = \begin{cases} 0.01 & : k = 0 \\ \gamma_{k-1} e^{-1/1000\mu} & : k > 0 \end{cases}. \tag{13}$$

Finally, the HC parameters are extracted from (7) using

$$\hat{K}_k = e^{\hat{\boldsymbol{\theta}}_{k(1)}}, \hat{B}_k = e^{\hat{\boldsymbol{\theta}}_{k(1)}}\hat{\boldsymbol{\theta}}_{k(2)}, \hat{n}_k = \hat{\boldsymbol{\theta}}_{k(3)}. \tag{14}$$

III. Proposed PI Methods

In this paper, we propose a TSPI method. In the first stage, which was originally presented in [34], a quadratic approximation of x^n is used to create the Quad-Poly HC model. The Quad-Poly parameters are identified using EWRLS. Quad-Poly can be used directly to predict the environment dynamics with a very high degree of accuracy. In the proposed second stage, TSPI, the nonlinear coefficient $\hat{n}$ is extracted from Quad-Poly in real-time using a lookup table and is used to linearly parameterize the HC (2). The coefficients $[\hat{K}, \hat{B}]$ are then estimated using the second-stage EWRLS. The block diagram of the proposed two-stage estimation process can be seen in Fig. 1. In contrast to polynomial models, which have previously been used to construct segmented models representing the dynamics of nonlinear objects [35], our method contains no discontinuities, and is sufficiently computationally efficient to run online.

A. Polynomial Approximation of x^n

The HC nonlinear coefficient n is known to be between 1 and 2 for most environments [16]; thus, x^n can be approximated using a Quad-Poly. Although a more accurate approximation can be calculated using least-squares (LS), the Taylor series expansion is useful to demonstrate how the polynomial coefficients behave. Using the binomial expansion, the standard Taylor series of the function $f(x)$ is rewritten in terms of increasing powers of x according to

$$f(x) = \sum_{l=0}^{\infty} \frac{f^{(l)}(a)}{l!} \sum_{k=0}^{l} \binom{l}{l-k} (-a)^{l-k} x^k \tag{15}$$

where $f^{(l)}(\cdot)$ represents the lth derivative of the function, and a is the center of the range of x, that is

$$a = \frac{1}{2}\sup[x(t)]. \tag{16}$$

In our case, $f(x) = x^n$. If n is not an integer, the lth derivative of x^n for $l = 1, 2, \ldots$ is given by

$$\frac{d^l}{dx^l}(x^n) = x^{n-l} \prod_{q=0}^{l-1} (n - q). \tag{17}$$

If n is an integer, the above equation only holds for all $l \leqslant n$. For x^n, the rearranged Taylor series is then

$$x^n = \sum_{l=0}^{\infty} \left[\frac{a^{n-l} \prod_{q=0}^{l-1}(n-q)}{l!} \sum_{k=0}^{l} \binom{l}{l-k} (-a)^{l-k} x^k \right]. \tag{18}$$

Equation (18) demonstrates that the coefficients of x^n depend not only on of n but also on the center value a and by consequence the maximum value of x. We truncate the Taylor Series to a quadratic ($l = 2$) polynomial, as a higher-order polynomial approximation will result in more terms to identify in EWRLS, and thus, require a richer excitation trajectory. For the robot maximum penetration (for example 1.5 cm in [20]), a more accurate approximation is found using LS

$$x^n \approx \alpha_1 x + \alpha_2 x^2. \tag{19}$$

Equation (19) must not include a constant term since this would result in a discontinuity at the point of contact ($x = 0$). A comparison between the truncated Taylor series polynomial (18) and the LS polynomial (19) for different values of n is shown in Fig. 2. The coefficient n in x^n can be found in reverse using the *identification ratio*

$$\delta = \alpha_1 / \alpha_2 \tag{20}$$

since n is unique for a given δ and the range of x. In order to identify $\hat{n}$ from $[\hat{\alpha}_1, \hat{\alpha}_2]$, a precomputed lookup table is required. The lookup table is calculated by identifying α_1 and α_2 from (19) for 1000 values of $1 \leq n \leq 2$ and the range of x that the robot is expected to travel. Next, δ is calculated for each n using (20). Using the lookup table, it is possible to work backward from $\hat{\delta}$ and estimate $\hat{n}$. The lookup table is shown graphically

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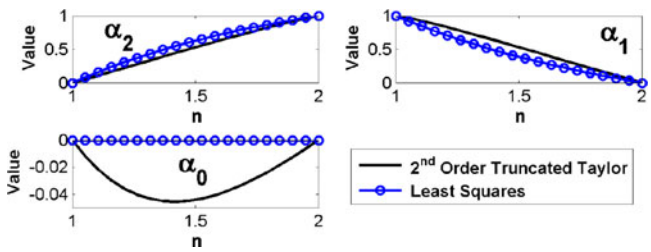


Fig. 2. Quad-Poly coefficients used to approximate x^n for the penetration range of $0 < x < 1$ cm and $a = 0.5$ cm, using different methods.

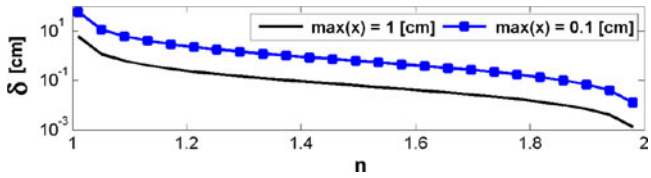


Fig. 3. Ratio δ calculated for two different maximum penetration estimates.

in Fig. 3. An example lookup table for $0 < x < 1.0$ cm is given in the matrix form

$$[\boldsymbol{n} \quad \boldsymbol{\delta}]^T = \begin{bmatrix} 1.0 & 1.1 & \dots & 1.9 & 2.0 \\ \infty & 5.534 & \dots & 0.068 & 0.000 \end{bmatrix}. \tag{21}$$

An upper-bound can be placed on the environment penetration (x) due to the physical size constraints of the robot and environment.

B. PI: Stage I

The environment dynamics are identified using a polynomial approximation of the HC model. To this purpose, a Quad-Poly model is developed by using the approximation (19) in the HC model (2), and is given by

$$F = K(\alpha_1 x + \alpha_2 x^2) + B(\alpha_1 x + \alpha_2 x^2)\dot{x} + \epsilon$$

$$= \boldsymbol{\omega}[x \quad x^2]^T + \boldsymbol{\beta}[x\dot{x} \quad x^2\dot{x}]^T + \epsilon \tag{22}$$

$$\boldsymbol{\omega} := [K\alpha_1 \quad K\alpha_2] \quad \boldsymbol{\beta} := [B\alpha_1 \quad B\alpha_2]. \tag{23}$$

As mentioned above, ϵ is the modeling and measurement error. The Quad-Poly model is linearly parametrized for EWRLS as

$$y_k = F_k = \boldsymbol{\phi}_k^T \boldsymbol{\theta}_k + \epsilon \tag{24}$$

$$\boldsymbol{\phi}_k^T = [x_k \quad x_k^2 \quad x_k\dot{x}_k \quad x_k^2\dot{x}_k] \tag{25}$$

$$\boldsymbol{\theta}_k = [\omega_{k1} \quad \omega_{k2} \quad \beta_{k1} \quad \beta_{k2}]^T \tag{26}$$

where the trivial initial values $\hat{\boldsymbol{\theta}}_0 = [0\ 0\ 0\ 0]$ or equivalently $[\hat{K}, \hat{B}] = [0, 0]$ have been chosen. For a discussion on initial parameters, see Section VI-D. Similarly to (18), the LS approximation (19) and by extension the polynomial model identified using EWRLS is dependent on the range of x considered. Advanced knowledge of the range is not required for Stage I of the identification; however, if the robot expands outside the expected range, EWRLS will naturally attempt to converge to a new model that covers the new range.

C. PI: Stage II

If the knowledge of the HC parameters $[\hat{K}, \hat{B}, \hat{n}]$ is required, they can be estimated using a lookup table (21) and a second-stage EWRLS. The δ ratios are dependent on the range of x, and therefore, the Stage II of identification requires advance knowledge of this range. As will be shown later, there is a large degree of leniency when estimating the range of x. Since the approximation (19) contains no dc offset term, the highest level of accuracy in most cases will be achieved if the excitation trajectory covers the *entire range* of x. It is sufficient to assume that the robot returns close to the environment boundary several times during the excitation.

1) Calculating Stiffness-to-Damping Ratio: Since $\hat{\boldsymbol{\omega}}$ and $\hat{\boldsymbol{\beta}}$ are not independent vectors, the coefficients $[\hat{K}, \hat{B}]$ cannot be analytically isolated, and only the stiffness-to-damping ratio can be calculated. This ratio

$$\gamma = \frac{K}{B} \approx \frac{|\hat{\boldsymbol{\omega}}|}{|\hat{\boldsymbol{\beta}}|} \tag{27}$$

may be useful to quickly determine the type of environment that the robot is in contact with.

2) Calculate $\hat{n}$: Theoretically, $\hat{\boldsymbol{\omega}}$ and $\hat{\boldsymbol{\beta}}$ can both be used to calculate the $\hat{\delta}$ ratio. In practice, $\hat{\boldsymbol{\omega}}$ yields better results since $\hat{\boldsymbol{\beta}}$ is directly dependent on a noisy velocity signal. For $1 \leqslant n \leqslant 2$, the LS approximation (19) will not involve negative $[\alpha_1, \alpha_2]$; however, noise, RLS errors, and modeling errors can cause the estimates to fall below zero. In order to avoid this problem, the absolute value of $\hat{\boldsymbol{\omega}}$ is used instead.

$$\hat{\delta} = \frac{|\hat{\omega}_1|}{|\hat{\omega}_2|} \approx \frac{\alpha_1}{\alpha_2}. \tag{28}$$

After each iteration of the EWRLS in stage 1, the $\hat{\delta}$ ratio is calculated from the estimated Quad-Poly model and compared to the lookup table (21) to find the closest match and the corresponding $\hat{n}$. As seen in Fig. 3, the δ ratios in the lookup table are naturally sorted, and thus, a binary search algorithm is used for improved computational efficiency. In this paper, a lookup table with 1000 values of n is used.

3) Second Stage EWRLS: The HC (2) can be linearly parameterized for K and B at each sampling period for a given $\hat{n}_k$ as

$$y_k = F_k = K_k x^{\hat{n}_k} + B_k x_k^{\hat{n}_k}\dot{x}_k \tag{29}$$

$$\boldsymbol{\phi}_k^T = [x_k^{\hat{n}_k} \quad x_k^{\hat{n}_k}\dot{x}_k], \quad \boldsymbol{\theta}_k = [K_k \quad B_k]^T. \tag{30}$$

The parameters K and B and, thus, the HC model can then be identified in the second stage using EWRLS. In this paper, the trivial initial values $[\hat{K}, \hat{B}] = [0, 0]$ have been chosen. The identified HC model is referred to as the TSPI model. It is important to note that unlike the method in [18], the first stage of the proposed method is independent of the second stage, and thus, it does not lead to the same performance limitations.

IV. Experimental Setup and Procedure

We experimentally evaluated our proposed PI methods Quad-Poly and TSPI against the benchmark LLI method for different environment types and conditions.

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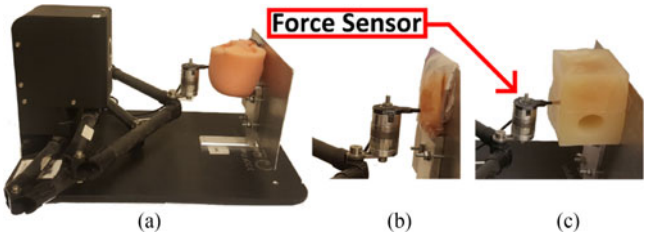


Fig. 4. Picture of the planar manipulator with the experimental environments. a) Foam. b) Chicken. c) PVC.

A. Experimental Setup

The experimental setup consists of a Quanser three degree-of-freedom (DOF) planar twin pantograph robotic manipulator and the environment under study. The three environments selected in order of increasing stiffness to damping ratio (γ) are i) sample of chicken breast tissue, ii) foam, and iii) polyvinyl chloride (PVC) phantom tissue. The chicken tissue was wrapped in a plastic bag for sanitation purposes. Fig. 4 shows the setup with the three environments. The manipulator is equipped with an ATI Nano-25 force/torque sensor and a 3 mm diameter cylindrical probe. The parameter n in the HC model is dependent on the geometry of the contact surface, and thus, the shape of the probe cross section. The selected probe was chosen to be small enough not to violate the point-contact assumption, yet large enough not to pierce the environment. The angle of rotation of each motor is measured by a high-resolution encoder with 20 000 counts/rev. The contact force is measured at a resolution of 1/48 N = 0.021 N. The robot is position-controlled to follow a desired line trajectory. The control system is implemented using MATLAB Coder Toolbox and Quanser QUARC real-time system operating at a sampling rate of 1 kHz on a 2.4 GHz Quad CPU.

B. Signal Conditioning

The robot position is available through the encoders and the velocity is calculated using Euler's backward-difference (BD) method. Since the velocity and force signals contain significant high-frequency noise, they need to be low-pass filtered. It is important to use the same filter for all signals in order to avoid phase offset errors. A first-order low-pass filter (LPF) with the cut-off frequency of $\omega_c = 50$ Hz is used. For velocity, position is low-pass filtered before BD. All filters used are infinite impulse response Butterworth filters as they are more computationally efficient than finite impulse response filters.

C. Units of Measurement

Unlike linear spring-damper models such as KV, the HC parameters $[K, B]$ cannot be linearly scaled when using different units of measurements. In this paper, centimeters and Newtons (cm, N) are used for all quantities. To understand the effects of nonlinearities, consider the nonrecursive LS case where the inverse of the regressor vector $\boldsymbol{\phi}^T = [\,\phi_1, \ldots, \phi_k\,]^T$ must be calculated. The condition number of $\boldsymbol{\phi}^T$, which reflects the sensitivity of LS to errors in $\boldsymbol{\phi}^T$, will be much larger for Quad-Poly if meters are used. For example in the trials with the chicken breast environment, the $\boldsymbol{\phi}^T$ condition numbers for Quad-Poly and LLI are 13 and 4.58, if centimeters are used, and 11 210 and 38 if meters are used, respectively. By rewriting the HC (2) in terms of cm/N

$$\begin{aligned} F &= K_{\text{cm}}(x_{\text{cm}})^n + B_{\text{cm}}(x_{\text{cm}})^n(\dot{x}_{\text{cm}}) \\ &= K_{\text{cm}}100^n(x_m)^n + B_{\text{cm}}100^{n+1}(x_m)^n(\dot{x}_m) \end{aligned} \tag{31}$$

the two following scaling rules are derived

$$K_m = 100^n K_{\text{cm}} \quad B_m = 100^{n+1} B_{\text{cm}} \tag{32}$$

$$\gamma_{\text{cm}} = \frac{K_{\text{cm}}}{B_{\text{cm}}} = 100\frac{K_m}{B_m} = 100\gamma_m\,. \tag{33}$$

D. Error Analysis

In this paper, the accuracy of the identified models is expressed in terms of the percent root mean squared error (%RMSE) defined as

$$\%\text{RMSE}(\hat{y}) = \frac{\sqrt{\frac{1}{N}\sum_{n=1}^{N}(\hat{y}[n] - y[n])^2}}{\sqrt{\frac{1}{N}\sum_{n=1}^{N} y[n]^2}} \times 100 \tag{34}$$

where $y[n]$ and $\hat{y}[n]$ denote the measured and predicted model output force, N is the number of samples considered, and n is the sample number.

E. Contact-Point and Scale Detection Tests

Accurate knowledge of the point of contact on the environment boundary (x_0) is required for parameter estimation, and is assumed to be constant in this paper. Furthermore, the knowledge of the maximum penetration depth, or the *scale* of the environment is also required. The above parameters are experimentally determined prior to running model identification experiments. To this end, the robot is commanded to advance in position control mode until a contact force is detected. In force control mode, the robot applies 2 N of force and records the scale. The robot then attempts to locate the point of contact with the environment by applying 0.05 N of force with quasi-static motion. This force level is about twice the force sensor minimum detectable force threshold. After the contact point and environment scale are determined, the lookup table is populated. As will be discussed in Section V-D, an exact maximum penetration estimate is not required, and the same lookup table can be used in the subsequent trials of the same environment. We would like to note that although visual observation indicated that the environments returned to their original shape after each trial, the point of contact was automatically re-estimated in each subsequent trial.

F. Cross-Validation Tests

Cross-validation tests using filtered-noise trajectories are used to determine the force prediction accuracy of the experimentally identified model. The cross-validation trajectory is designed to have similar amplitude and frequency characteristics as the identification trajectory. The original velocity signal used for EWRLS is noisy and is passed through a fourth-order

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TABLE I
CROSS-VALIDATION %RMSE OF HC MODEL FOR CHICKEN

	Discontinuous		Continuous	
Method	Online	Offline	Online	Offline
Quad-Poly	6.7%	7.9%	8.1%	9.1%
TSPI	9.0%	9.5%	13.0%	11.3%
LLI	15.2%	18.4%	12.0%	14.3%
KV	26.0%	26.0%	26.9%	26.8%

LPF with $\omega_c = 12.5$ Hz. The additional filtering is applied only to the velocity signal to minimize the phase offset between the measured and predicted force. To compare the predicted force and the noise-free measured force, the measured force signal is passed through a noncausal, zero-phase fifth-order LPF with $\omega_c = 5$ Hz. The zero-phase force signal is then compared to the force predicted online by the model. Since the model identification and cross validation are performed in separate trials, any material conditioning, which occurred during the two trials is reflected in the cross-validation error.

V. EXPERIMENTS AND RESULTS

A. Model Identification Experiments and Validation

The main HC model identification experiments were conducted on the three mentioned environments under *continuous* and *intermittent* robot-environment contact conditions in order to assess the suitability of KV versus HC and compare the performance of the three HC identification methods, Quad-Poly, TSPI, and LLI.

The robot was instructed to move along an excitation trajectory for 10 s, and EWRLS would start once $x > x_0$. The unitless normalized desired trajectory is given by

$$x_{dn}(t) = G[0.5\sin(0.6\pi t) + 0.3\sin(1.6\pi t) + 0.1\sin(3.52\pi t) + 0.1\sin(4\pi t)] - D \quad (35)$$

where the condition $G + D = 1$ is enforced to ensure that the maximum penetration is equal to 1. The frequency components are selected such that there are no integer multiple frequencies. In order to ensure that the identified dynamics reflect the dynamic behavior of the entire environment, the normalized trajectory $x(t)$ was scaled to match the environment maximum penetration estimate (described in Section IV-E). A bias term was added to realize the intermittent contact condition. *Note:* The discrete-time index k is not incremented when contact is lost, thereby freezing the parameter estimates.

Each experiment was repeated three times for each testing condition and the results reported in Tables I, II, and III reflect the mean value of the force prediction errors in %RMSE. The results under the *Online* column represent the model identified online using EWRLS, whereas the results under the *Offline* column evaluate the model obtained offline using batch LS. Batch estimation does not require setting initial values and tuning the adaptation gain (12), and it places equal weight on the entire collected data. For offline assessment, the position and force signals were passed through a noncausal zero-phase fifth-order LPF with $\omega_c = 5$ Hz, and only one $\hat{n}$ extraction was required for TSPI.

TABLE II
CROSS-VALIDATION %RMSE OF HC MODEL FOR FOAM

	Discontinuous		Continuous	
Method	Online	Offline	Online	Offline
Quad-Poly	6.5%	6.7%	8.1%	8.8%
TSPI	6.1%	6.4%	8.3%	8.2%
LLI	17.7%	16.0%	10.4%	11.2%
KV	16.1%	15.8%	15.6%	15.7%

TABLE III
CROSS-VALIDATION %RMSE OF HC MODEL FOR PVC

	Discontinuous		Continuous	
Method	Online	Offline	Online	Offline
Quad-Poly	3.6%	3.6%	5.1%	5.0%
TSPI	3.9%	5.8%	5.5%	6.6%
LLI	10.5%	10.4%	6.1%	5.9%
KV	12.2%	12.1%	13.3%	12.9%

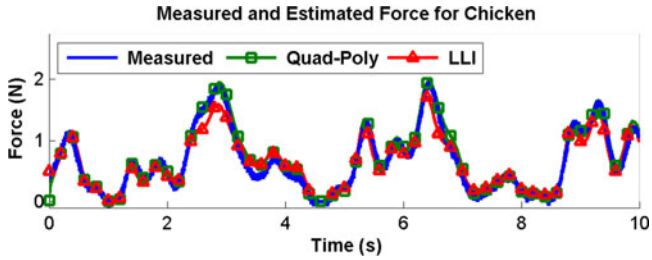


Fig. 5. Measured and real-time estimated force for a single intermittent contact trial with the chicken environment.

The high %RMSE for KV confirms that it is not suitable for soft environments. Among the HC models, Quad-Poly, and TSPI perform well under continuous and intermittent contact. On average TSPI is 1.2% less accurate than Quad-Poly. This decrease is caused by the propagation of EWRLS errors, $\hat{n}$ extraction errors, and modeling errors. The advantage of Quad-Poly is most distinct in the chicken breast environment, likely due to the fact that it is a highly nonhomogeneous environment. The increase in accuracy in discontinuous contact trials is attributed to the trajectory more fully covering the entire range of x. LLI performed poorly compared to Quad-Poly under all test conditions and compared to TSPI under all conditions except for online continuous contact with chicken. In order to visualize the difference in accuracy between PI and LLI, the measured and the force estimated online for a single trial in chicken under discontinuous condition are shown in Fig. 5. The force prediction improvement of Quad-Poly and TSPI are important since prediction errors cause poor performance in robotic systems [9], [26].

From (3) and (4), the LLI performance is expected to degrade for highly damped environments and for excitation trajectories, which include the area close to the environment contact point. These expectations are confirmed experimentally

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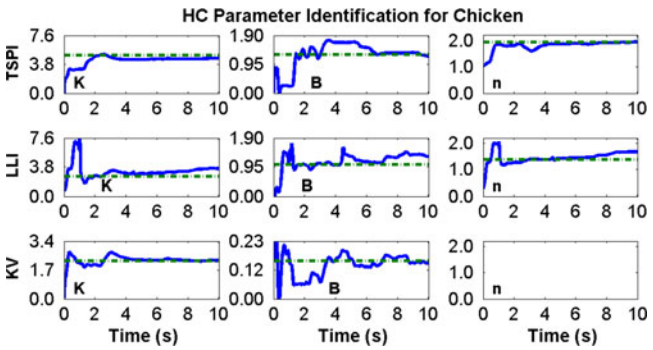


Fig. 6. Real-time estimation of the HC parameters for the chicken breast tissue environment. Offline estimates shown as dashed lines.

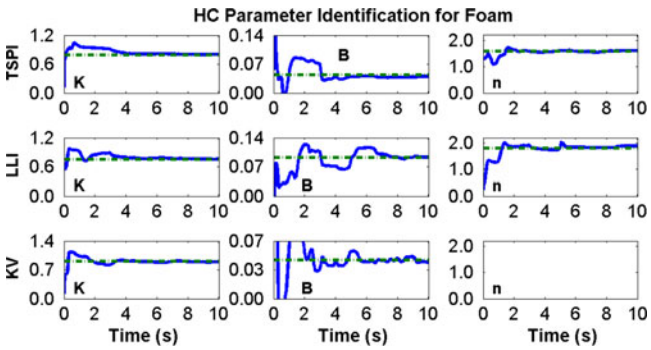


Fig. 7. Real-time estimation of the HC parameters for the foam environment. Offline estimates shown as dashed lines.

since the %RMSE of LLI was over 2 times higher than Quad-Poly for discontinuous contact. As well, LLI was only 1.0% less accurate for continuous contact in the PVC environment, which has the highest stiffness to damping ratio. It is important to note that for continuous contact trials, LLI recursive identification must start after the robot made contact to avoid large errors (this is not necessary for Quad-Poly/TSPI). Since guaranteeing contact at all times is not always desirable or practical, the remainder of the paper considers only intermittent contact.

B. Parameter Estimation

Real-time parameter estimation is performed using LLI and TSPI over an interval of 10 s. The *percent standard deviation* $c[k]$ is a discrete function of time used to gauge the convergence of the estimate, and is given by

$$c[k] = \frac{\sigma[k, k+1000]}{\text{rms}[k, k+1000]} \times 100 \tag{36}$$

where $\sigma[k, k+1000]$ and $\text{rms}[k, k+1000]$ are the standard deviation and RMS of the estimate over the next 1000 samples (i.e., 1 s) after point k. The "estimation convergence time" k_c is defined as the point at which the following two conditions are satisfied:

$$\sup\left[c[k] : 0 < k < k_c\right] > 5\% \tag{37}$$

$$\sup\left[c[k] : k_c < k < k_e\right] < 5\% \tag{38}$$

where k_e is the end point of the estimation. The *percent standard deviation after convergence* is, thus, given by

$$c_\infty = \frac{\sigma[k_c, k_e]}{\text{rms}[k_c, k_e]} \times 100. \tag{39}$$

The final value of the estimate is taken as the mean value of the last second (last 10% of the estimates). The HC parameter estimates from a single trial for the chicken breast tissue, foam and PVC are illustrated in Figs. 6, 7, and 8 where the online estimates are compared to the offline LS estimates. The final values, standard deviation after convergence, and convergence times over all trials are also provided in Tables IV, V, and VI.

Compared to LLI, the online TSPI of $[\hat{K}, \hat{n}]$ exhibit smaller fluctuations at the beginning of the estimation, especially for the chicken breast environment in Fig. 6. One possible explanation is that at the onset of penetration where $x(t)$ is small, the condition (4) for LLI is violated. Furthermore, TSPI estimates of $[\hat{K}, \hat{B}, \hat{n}]$ have a smaller percent standard deviation and take less time to converge. The mean convergence times reported for B for the LLI and KV methods, which are over 9 s indicate that in the parameter did not converge within the 10 s period of the experiment.

As can be seen in Figs. 6–9 as well as Tables IV–VI, the damping parameter $\hat{B}$ (or $[\hat{\beta}_1, \hat{\beta}_2]$) estimates take the longest to converge for all methods. TSPI, LLI, and KV all exhibit sudden

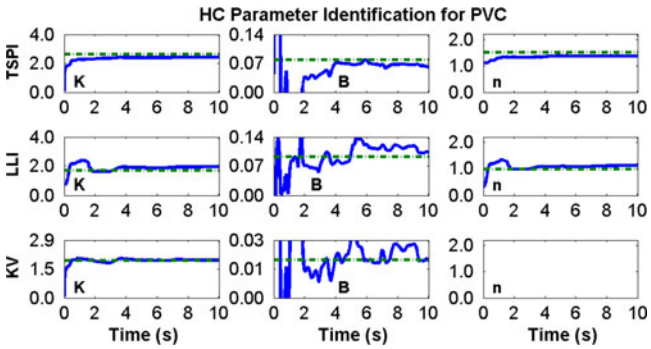


Fig. 8. Real-time estimation of the HC parameters for the PVC environment. Offline estimates shown as dashed lines.

TABLE IV
HC PARAMETER ESTIMATION STATISTICS FOR CHICKEN

	Final Value (% Std Dev after Conv)			Conv. Time (s)		
Method	K	B	n	K	B	n
TSPI	4.11 (3.1)	1.09 (3.6)	1.95 (3.5)	1.6	6.1	2.0
LLI	3.29 (7.2)	1.20 (3.7)	1.73 (8.8)	2.6	7.1	1.5
KV	2.05 (5.0)	0.136 (4.3)	N/A	2.6	8.5	N/A

TABLE V
HC PARAMETER ESTIMATION STATISTICS FOR FOAM

	Final Value (% Std Dev after Conv)			Conv. Time (s)		
Method	K	B	n	K	B	n
TSPI	0.813 (6.5)	0.0387 (3.9)	1.61 (2.3)	0.1	4.5	1.3
LLI	0.768 (4.5)	0.0939 (5.8)	1.87 (2.7)	1.5	6.4	3.4
KV	0.899 (2.5)	0.0439 (4.9)	N/A	2.4	8.9	N/A

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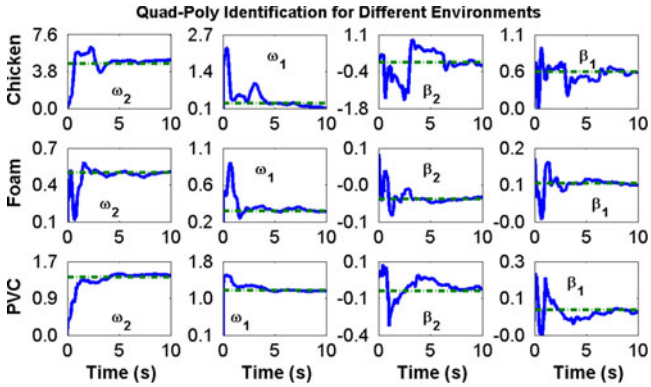


Fig. 9. Real-time estimation of the Quad-Poly model for different environments. Offline estimates shown as dashed lines. Units: ω_1 (N/cm), ω_2 (N/cm^2), β_1 (Ns/cm), β_2 (Ns/cm^2).

TABLE VI
HC PARAMETER ESTIMATION STATISTICS FOR PVC

	Final Value (% Std. Dev after Conv)			Conv. Time (s)		
Method	K	B	n	K	B	n
TSPI	2.44 (2.9)	0.0611 (4.0)	1.39 (3.1)	0.4	7.3	0.3
LLI	1.94 (2.7)	0.0987 (5.0)	1.12 (4.1)	3.1	9.1	1.6
KV	1.90 (2.9)	0.0205 (5.0)	N/A	0.3	9.3	N/A

fluctuations for $\hat{B}$. The challenge in estimating the environment damping can be attributed to the quality of the velocity signal. The velocity signal is heavily quantized and often inaccurate as it is approximated from the measure position signal. Furthermore, the richness of the velocity signal is limited by the excitation trajectory. Slow convergence may limit performance in applications such as adaptive control during intermittent contact tasks (e.g., robotic suturing) since $\hat{B}$ will not converge during one contact period. However, freezing the parameter estimates when contact is lost can partially mitigate this problem.

In conjunction with %RMSE data reported in Tables I, II, and III, it can be seen that the KV error increases for environments with larger values of $\hat{n}$ in their HC models. Quad-Poly parameters do not directly represent K, B, and n. However, since they are used in TSPI to compute the HC parameters, samples of their estimate profiles are also shown in Fig. 9. The parameters converge to their offline estimates within the 10 s interval.

As mentioned above, online applications benefit from accurate estimates of the HC model. In this context, model convergence is studied in terms of the force prediction accuracy of the *transient* HC model. This is done by taking snapshots of the estimated model in 0.01 s intervals, and using the transient model to predict the force for the *entire* trajectory. As seen in Fig. 10, not only are Quad-Poly and TSPI significantly more accurate in steady state, but their transient error diminishes faster than LLI for all three environments. Due to the nonhomogeneous nature of the chicken environment, the transient error takes much longer to dissipate for all methods. In addition, as the robot breaks contact with the environment between 1 s and 2 s, the LLI prediction error shoots up sharply for the foam and PVC environments.

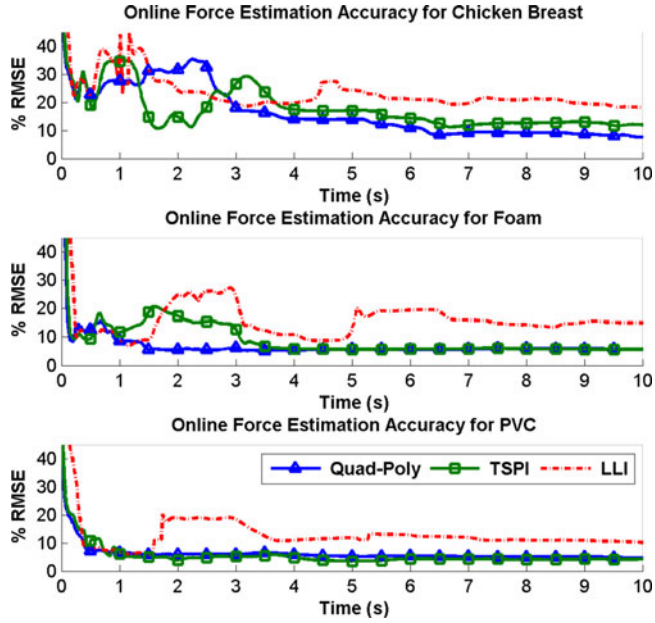


Fig. 10. Accuracy of online force estimation as a function of time for the three environments.

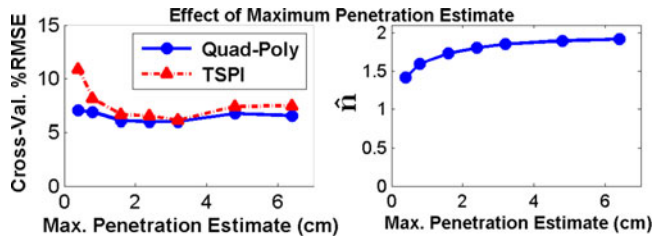


Fig. 11. Second stage EWRLS provides adequate robustness to changes in $\hat{n}$ and the accuracy of TSPI model is maintained for larger penetration estimates.

C. Effect of Maximum Penetration Estimate

The requirement that the maximum penetration depth, $x_{\max}$, be known prior to constructing the lookup table is a potential drawback of TSPI. In order to study the effect of the estimate $\hat{x}_{\max}$, additional trials were performed using the foam environment where $\hat{x}_{\max}$ was varied from the measured value of 1.6 cm over a range of 0.4–6.4 cm. The experiment was repeated 10 times for each value of $\hat{x}_{\max}$, and the effect on cross-validation %RMSE and $\hat{n}$ are shown in Fig. 11. The accuracy of Quad-Poly is ideally independent of $\hat{x}_{\max}$ and serves as a benchmark for the TSPI identification accuracy. The second stage of TSPI can account for small changes in $\hat{n}$. The change in $\hat{n}$ and %RMSE is relatively small for penetration estimates beyond 1.6 cm. Therefore, for TSPI, it is advisable to overestimate $\hat{x}_{\max}$ than to underestimate. If $\hat{x}_{\max}$ cannot be found experimentally, as has been done in the paper, then the order of maximum penetration can be approximated. The maximum penetration depth is assumed to be limited by either the maximum reach of the robot x_r, or by F_m/K_r where F_m is the maximum force exerted by the robot and K_r is a rough estimate of the order of environment stiffness. As discussed, an approximate overestimation is sufficient.

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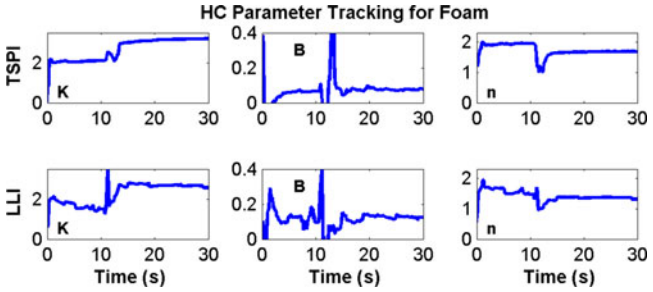


Fig. 12. HC model parameters estimates when the robot switches to dynamically different environment.

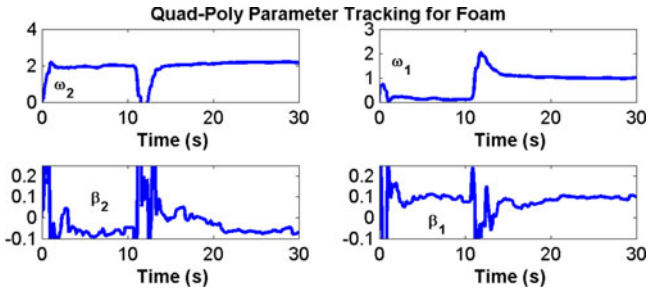


Fig. 13. Quad-Poly parameter estimates when the robot switches to dynamically different environment.

D. Parameter Tracking for Varying Environments

If the EWRLS forgetting factor λ (12) is held constant at some value less than 1, EWRLS can adapt to environment time-varying parameters. However, this requires careful tuning as the steady-state estimates will be less stable. Self-perturbing-RLS, which adds a perturbing term to the RLS covariance matrix $\boldsymbol{P}$ (9) if large errors are detected, was used in [28]; however, in our experiments the results were not satisfactory. Instead, λ is reset to its original value when the *a posteriori* force estimation error goes beyond the predefined threshold E_{ft}, as shown in Fig. 1. The adaption gain γ_k, introduced in (13), is, thus, adjusted according to

$$\gamma_k = \begin{cases} 0.01 & : k = 0 \\ 0.01 & : |F - \hat{F}| > E_{ft} \\ \gamma_{k-1} e^{-1/1000\,\mu} & : \text{otherwise.} \end{cases} \tag{40}$$

The threshold is set to detect the prediction error caused by significant change or error in the parameter estimates. Our experiments showed that the threshold $E_{ft} = 0.4$ N (or 20% of full-scale) captured these changes while minimizing the amount of "false positives" during continuous operation for the environment studied in this section.

The parameter tracking capabilities of PI and LLI were investigated by switching between two environments: 1) soft foam on top of hard PVC, and 2) soft PVC. These two environments were selected since they have drastically different K and n parameters. The same identification procedure as before was used, however, after 15 s in the first environment, the robot moves to the second environment for a further 15 s. The parameter estimation results for HC models and Quad-Poly are shown in Figs. 12 and 13. By resetting the adaptation gain, all three methods were able to track the change and reach a new stable estimate. The transient error of the estimation algorithm should be as small as possible and not produce large spikes. TSPI produced spikes for $\hat{B}$ only; however, LLI produced spikes for $\hat{K}$ and $\hat{B}$. After the transient error subsided, TSPI demonstrated a clearer distinction between the two environments. The Quad-Poly model displays spikes only in $\hat{\boldsymbol{\beta}}$.

E. Manual Exploration

The purpose of this section is to evaluate the performance of the HC estimator when the excitation trajectory is generated by a human operator instead of by the robot following the desired trajectory (35). This is referred to as "manual exploration" of the environment. This analysis was motivated by previous uses of LLI in identifying dimensional HC model of an object with unknown geometry for use in virtual environments [36]. An operator can easily accomplish this task by manually moving the force probe at various angles and penetrations. A second motivation is online identification for robotic contact tasks using nonplanned, human-generated trajectories, e.g., remote control of the da Vinci surgical robot. Manual exploration experiments were conducted using the foam environment. The operator freely moves the force probe in the x direction, while the other DOFs are constrained using tight potential differenc control.

In order to estimate the contact point x_0, the operator is instructed to repeatedly "tap" the environment in quasi-static motion. When a contact force is first detected, the position is registered and labeled as x_c. It is assumed that probe-environment contact happens before contact force is sensed, i.e., $x_0 < x_c$ for two reasons. First, due to the sensor's noise floor, a contact force cannot be definitively registered in a deadband close to the environment boundary. Second, due to the delay caused by the force sensor filter, the contact force is not detected immediately. The filtering delay of a few milliseconds will have a negligible effect when the operator is moving at low velocities. However, its effect would be significant at a relatively high speed, e.g., 30–50 cm/s.

After registering x_c, the contact point estimate $\hat{x}_0$ needs to be found. Thus, 40 values of $\hat{x}_0$ equally spaced by 0.5 mm are selected prior to x_c. For each $\hat{x}_0$, the penetration trajectory

$$x'[k, \hat{x}_0] = \begin{cases} x[k] - \hat{x}_0 & : x[k] \geqslant \hat{x}_0 \\ 0 & : x[k] < \hat{x}_0 \end{cases} \tag{41}$$

is created. The operator is the instructed to make small environment penetrations by exercising quasi-static motion. The measured force and the 40 penetration trajectories are fit to 40 linear spring models $F = Kx'$ using batch LS with a sliding window of 2.0 s. The value of $\hat{x}_0$, which results in the lowest predicted force %RMSE is chosen as the true value.

As will be seen in Section VI-A, the resolution of the chosen contact point candidates $\hat{x}_0$, that is 0.5 mm, is set to capture the minimum model force prediction error. After determining the contact point, the operator moves the probe to match a force profile consisting of three repeating one-second intervals: high

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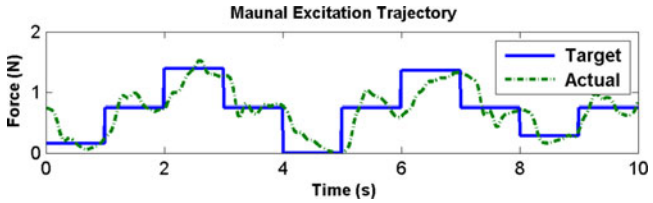


Fig. 14. User is asked to follow a randomly generated target excitation trajectory consisting of one second high, mid, and low penetration stages.

TABLE VII
CROSS-VALIDATION FOR MANUAL EXPLORATION OF FOAM

Method	Online	Offline
Quad-Poly	8.9%	10.6%
TSPI	8.9%	10.7%
LLI	14.4%	24.9%
KV	20.7%	19.9%

penetration, mid-penetration, and low penetration, as shown in Fig. 14. The mid-penetration point is fixed to 0.75 N, and the high and low penetrations are uniform random plateaus in the ranges of [1.25 N, 2 N] and [0 N, 0.5 N]. This profiles allows for random selection of the environment interior points.

In order to ensure that the results were repeatable given an unpredictable operator, the experiment was repeated 5 times. The cross-validation trial used the same filtered noise excitation trajectory as before to reduce the variability in the experiments. It was observed that in order to develop a model that can accurately predict the force in a cross validation trial, the excitation trajectory in both trials needed to have similar amplitude and velocity characteristics. Furthermore, the mean penetration value of both trajectories must also be similar. As seen in Table VII, Quad-Poly and TSPI are significantly more accurate than LLI for manual exploration. However, they are less accurate than the automated exploration trials reported in Table V due to the richness of the trajectory.

VI. ANALYSIS OF EXPERIMENTAL CONDITIONS

In this section, we investigate the effect of variations or uncertainties in some system parameters or collected data on the estimation accuracy. Offline simulations were conducted using the same EWRLS identification algorithms (Quad-Poly, TSPI, LLI) and previously collected experimental data from five foam environment trials. These studies introduce an error or parameter change in the identification trial, and compute the cross-validation accuracy in a second trial *without* the change.

A. Effect of Environment Contact Point

The contact point estimate $\hat{x}_0$ significantly affects the accuracy of the model, although this effect is limited for the soft environments in this paper. The $\hat{x}_0$ error causes the system to converge to a different LS solution, thus, introducing error when the model is applied to the cross-validation trial without a $\hat{x}_0$ error. As seen in Fig. 15(a), a ± 1 mm $\hat{x}_0$ error during the identification doubles the cross-validation error. Although LLI produces a larger force prediction error, it demonstrated better robustness to $\hat{x}_0$ error, in part since LLI takes the natural

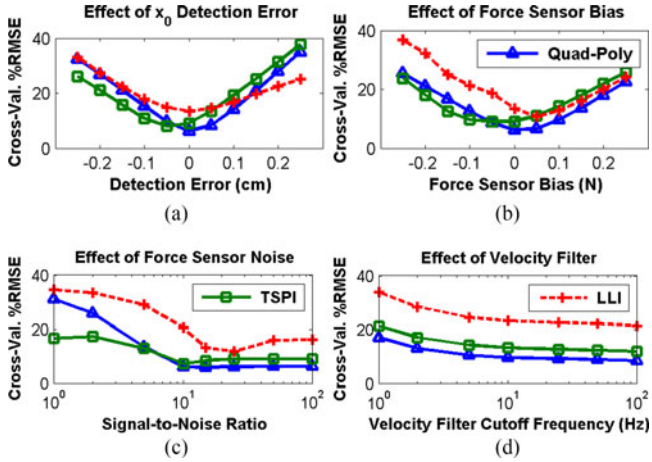


Fig. 15. Effect of a) x_0 error, b) force sensor bias, c) force sensor noise, and d) velocity filter on cross-validation RMSE.

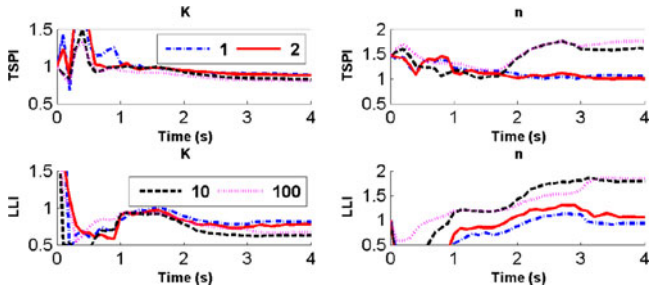


Fig. 16. Effect of the force noise SNR on the transient estimation of K an n for TSPI and LLI for a single trial in the foam environment.

logarithm of x whereas Quad-Poly takes the square, which amplified the effect of x_0. Note that for stiff environments where F changes significantly in one encoder tick, force-only estimation may be more appropriate [4], [5].

B. Effect of Force Noise and Bias

LS algorithms naturally provide a certain robustness to sensor noise but not to force sensor bias, both of which are present in practical scenarios. To investigate, we separately introduced bias and altered the signal-to-noise ratio (SNR) in the measured force. The force sensor bias causes a similar problem to the $\hat{x}_0$ error, forcing EWRLS to a different LS solution. As shown in Fig. 15(b), the force sensor bias should be kept below 0.1 N, or 5% of full-scale, in order to maintain good accuracy. Figs. 15(c) and 16 demonstrate the effect of noise on prediction error and the estimated parameters *K* and *n*. It is observed that an SNR of at least 10–15 db is required for each method to predict contact force with its highest possible accuracy. The increase in error for low SNR can be attributed to the error in the estimation of *n*, which drops from 1.7 for high SNR to about 1 for low SNR. Above this SNR level, LS is able to sufficiently cancel the effect of noise. It is notable that Quad-Poly and TSPI methods better maintain their accuracy for lower SNR than did LLI.

C. Effect Velocity Filter

There is a tradeoff in choosing the velocity filter cutoff frequency when estimating the velocity from the measured

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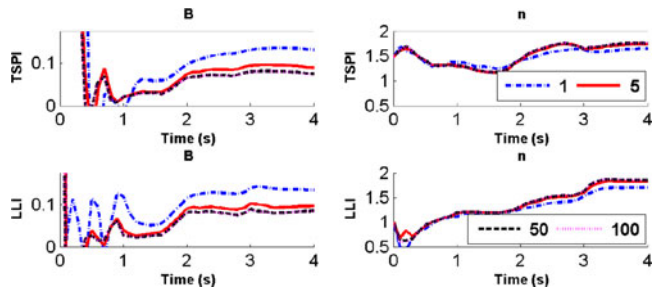


Fig. 17. Effect of the velocity filter cutoff frequency on the transient estimation of B an n for TSPI and LLI for a single trial in the foam environment.

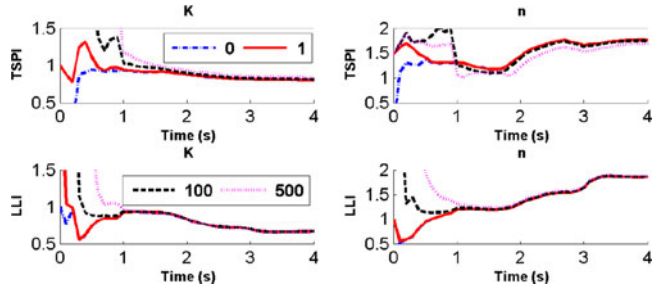


Fig. 18. Effect of initial parameters θ_0 on the transient estimation in the foam environment for a single trial.

position. High cutoff frequency results in increased noise, whereas low cutoff frequency suppress the velocity content. Figs. 15(d) and 17 illustrate the effect of filter frequency on the prediction error and parameter estimation. Reducing the cutoff frequency down to 1 Hz, which is below the maximum excitation frequency of 2 Hz, results in velocity suppression and large phase offset, and hence, a significant loss of accuracy. This adverse effect is also visible in the identified B, and to a lesser degree in n and K (not shown here due to space limitation). It is also determined that no noise issue is present for cutoff frequencies above our nominal value of 50 Hz, as the LS is by nature noise resistant. It is, thus, recommended to use cutoff frequencies well above the excitation frequency.

D. Effect of Parameter Initialization

Fig. 18 illustrates the effect of initialization on the identified parameters. Here the initial values are chosen according $\boldsymbol{\theta}_0 = N[1\ 1\ 1], N = 0, 1, 100, 500$. As expected, the HC parameters K and B (and similarly the Quad-Poly parameters ω and β) are obtained from linear LS identification, and thus, the final identified values are independent of the parameter initialization. However, initialization values vary the transient regime or to a lesser degree the convergence time. In general, LLI produces greater initial overshoot in the estimated parameters, especially when initial values become further distant from the final values as N goes to 500.

VII. Conclusion

A novel PI algorithm, motivated by the need to develop accurate online dynamic models for soft environments with limited deformation at a single-point of contact was presented. The TSPI algorithm generates a Quad-Poly approximation of the HC dynamic model in the first stage. In the second stage, it identifies the HC model parameters using a lookup table and RLS. The algorithm is computationally efficient and can run online with a 1 kHz sampling frequency. The identification procedure, including contact point detection and environment scaling were discussed.

The Quad-Poly and TSPI methods were experimentally assessed against the benchmark LLI method. TSPI was on average 1.2% less accurate than Quad-Poly due to EWRLS propagation errors, however, both TSPI and Quad-Poly were shown to be a significant improvement over LLI for estimating the environment HC model. This is especially true in the case of discontinuous contact with the environment, where the cross-validation error of LLI was over 2 times that of Quad-Poly for all three environments tested. In most cases, the Quad-Poly and TSPI models converge faster than the LLI model, demonstrate superior parameter tracking capability, and perform well when the environment is probed manually. However, more work is required to improve the damping estimate convergence time.

Additional analysis was performed on the robustness of Quad-Poly and TSPI. Contact point estimation error and force sensor bias were shown to have significant effects on cross-validation error, however, the force sensor noise and velocity filtering had a minimal impact. Expansion to segmented HC models for non-homogeneous environments is a possible further research topic. The HC model is increasingly being adopted for modeling soft tissues and designing controllers for robotic interaction with such environments with limited deformation at a single-point of contact. The improved accuracy, convergence time and robustness of Quad-Poly, and TSPI will be valuable for further development of these applications and for the integration of HC models and FEM for a broader class of deformable soft tissues models.

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